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Mathematical model analysis and solution properties of nonlinear filtration processes in multidimensional domains

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ABSTRACT

This paper investigates a nonlinear thermo-chemical diffusion-reaction system characterized by coupled parabolic equations that govern the spatio-temporal evolution of temperature and concentration-dependent processes. The model incorporates nonlinear diffusion coefficients, cross-coupling terms, and nonlinear source functions, which collectively describe the complex interplay between heat and mass transfer in reactive media. A robust numerical framework is developed based on an Alternating Direction Implicit (ADI) scheme of the Peaceman-Rachford type, allowing for efficient and stable time integration of the nonlinear system. The implementation ensures high computational efficiency and improved numerical stability, particularly for stiff reaction terms and strongly coupled diffusion dynamics. Comprehensive numerical experiments are conducted to validate the accuracy and stability of the proposed scheme. Error convergence analysis confirms the expected second-order spatial and first-order temporal accuracy, while the time-step sensitivity and stability tests demonstrate the robustness of the algorithm under various discretization parameters. Boundary layer behavior is also examined to capture localized gradients and nonlinear interaction patterns. The obtained results reveal that the proposed computational framework accurately reproduces the characteristic thermo-chemical diffusion phenomena and maintains stability even under extreme parameter regimes. The study provides a reliable numerical tool for analyzing multi-scale diffusion-reaction systems relevant to chemical engineering, materials processing, and thermal energy storage applications.

1. Introduction

Nonlinear thermo-reactive diffusion processes play a crucial role in many physical and engineering systems, including heat-assisted chemical reactions, material synthesis, and thermal transport in heterogeneous media. The interaction between diffusion and thermal effects often leads to complex spatiotemporal behavior that can only be captured through coupled nonlinear partial differential equations.

In this study, we consider a nonlinear thermo-reactive diffusion model describing the evolution of two interacting quantities governed

by temperature-dependent diffusivity and reactive source terms. The system incorporates nonlinearities of power-law type and cross-interaction effects that represent realistic physical coupling mechanisms observed in thermochemical systems.

The primary aim of this work is to develop a stable and accurate numerical framework for solving this class of coupled nonlinear diffusion-reaction systems using the alternating direction implicit (ADI) method combined with the Thomas algorithm for efficient computation. The obtained numerical results provide insights into the spatial evolution, boundary layer structures, and long-term stability of the solution

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under physically relevant parameter regimes.

Background and Motivation: Thermo-reactive diffusion models serve as a cornerstone in the analysis of heat and mass transfer in reactive media, such as catalytic reactors, thermal oxidation of alloys, and chemical vapor deposition processes. These models capture the nonlinear coupling between temperature-dependent diffusion and reaction kinetics, which are essential for predicting real-world phenomena in materials science, energy systems, and environmental engineering.

Recent studies (e.g., Wang [1], Li [2]) have demonstrated that nonlinear diffusion with variable coefficients can generate spatial instabilities and pattern formation under certain thermal conditions. However, most classical formulations either neglect the cross-reaction effects or assume constant diffusivity, leading to reduced accuracy when applied to thermally active materials.

In the context of Uzbekistan's industrial and environmental systems, such models are particularly relevant for analyzing thermal diffusion in chemical reactors, reactive transport in underground aquifers, and heat-induced transformations in composite materials used in energy storage technologies. For instance, in localized thermochemical processing of natural mineral composites, diffusion and reaction mechanisms strongly depend on temperature gradients, which aligns with the nonlinear coupling represented in our model.

Motivated by these challenges, this work focuses on developing a robust computational approach to simulate such complex thermo-reactive diffusion systems efficiently. The proposed framework allows accurate representation of nonlinear diffusion behavior, cross-reaction feedbacks, and the influence of boundary conditions - providing a deeper understanding of the processes observed in both global thermochemical systems and local industrial applications in Uzbekistan.

1.1. Research gap and limitations of existing models

Despite the significant progress in modeling thermo-reactive diffusion processes, several important limitations remain in the existing literature. Traditional models of nonlinear diffusion often assume constant thermal conductivity or homogeneous reaction rates, which fail to capture the real behavior of thermally sensitive materials. In particular, most diffusion-reaction systems neglect spatially varying diffusivity coefficients and cross-coupled reactive feedbacks, both of which play a crucial role in accurately describing non-equilibrium processes in thermo-chemical environments (Rohde [3], Payne [4], Gurtin [5], Chen [6]).

Moreover, earlier studies on thermochemical diffusion primarily focused on one-dimensional or steady-state approximations, limiting their applicability to complex multi-dimensional geometries. These simplifications tend to suppress critical behaviors such as boundary layer formation, nonlinear interaction of diffusion fronts, and transient blow-up phenomena. Consequently, the predictive capability of such models becomes insufficient for high-temperature and multi-species diffusion systems observed in industrial and environmental settings (Shmarev [7]).

Another major gap lies in the numerical treatment of these systems. Many existing algorithms rely on explicit schemes that are conditionally stable or computationally expensive for stiff nonlinear systems. The absence of efficient alternating-direction and implicit-explicit hybrid schemes limits the ability to simulate long-term behavior under realistic thermal gradients. Furthermore, validation studies often lack quantitative error analysis or comparison against reference solutions, which weakens the practical reliability of these models.

Therefore, there exists a strong need to formulate and analyze a generalized thermo-reactive diffusion system-such as the one presented in this study-where both diffusion and reaction mechanisms are nonlinear, interdependent, and spatially variable. Addressing these gaps enables the construction of more accurate and stable computational frameworks suitable for realistic thermo-chemical processes encountered in materials science and energy technologies.

1.2. Main contributions

The present study develops and analyzes a generalized nonlinear thermo-reactive diffusion model that extends classical heat and mass transfer theories to more realistic thermo-chemical environments. The main contributions of this work can be summarized as follows:

1. Formulation of a coupled nonlinear diffusion-reaction system. A new mathematical model is proposed in the form of system (1)-(3), incorporating temperature-dependent diffusivity coefficients, nonlinear reaction kinetics, and mutual coupling effects between reactive species. This formulation generalizes existing diffusion models by allowing variable thermal and chemical conductivity functions $\sigma_1(\chi)$, $\sigma_2(\chi)$, $\sigma_3(\chi)$.
2. Theoretical results on existence and stability of weak solutions. Using energy estimate techniques and compactness arguments, we establish the existence, uniqueness, and boundedness of weak solutions under realistic assumptions on the coefficients. These results provide a rigorous mathematical foundation for the proposed system and ensure the physical well-posedness of the model.
3. Development of efficient numerical algorithms. An alternating-direction implicit (ADI) scheme combined with the Peaceman-Rachford approach and Thomas algorithm is designed to solve the nonlinear system with improved stability and computational efficiency. The scheme is further validated by error convergence and time-step sensitivity analyses, demonstrating high numerical accuracy.
4. Quantitative and physical interpretation of thermo-reactive diffusion. The study provides 3D simulations and contour visualizations of thermal and concentration distributions under realistic conditions. These numerical experiments reveal the interaction between thermal gradients and reactive diffusion fronts, illustrating how nonlinear coupling affects boundary layer formation and steady-state behavior in thermochemical systems relevant to Uzbek industrial materials and global thermal coating processes.
5. Framework for future applications. The proposed formulation and methodology can be directly extended to more complex thermo-mechanical systems, including multi-component alloys, reactive porous media, and energy storage materials, providing a universal basis for modeling coupled transport phenomena in modern material technologies.

1.3. Related work

The study of nonlinear thermo-reactive diffusion processes has attracted substantial attention across physics, materials science, and applied mathematics. Classical models-such as Fick's diffusion law and Fourier-type heat conduction-provide fundamental descriptions of transport phenomena but fail to account for strong nonlinearities and thermally coupled reaction dynamics observed in modern engineering materials.

In early works, Kaviani [8] and Gurtin [5] examined heat and mass transfer in reactive media, emphasizing temperature-dependent diffusivities but assuming spatial homogeneity. Later, Volpert [9] extended the framework to nonlinear reaction-diffusion systems in heterogeneous media, revealing pattern formation and blow-up regimes. Similarly, Shigesada [10] and Konkoli [11] analyzed cross-diffusion models in population dynamics and porous media, laying the theoretical basis for coupled diffusion mechanisms.

However, most existing studies neglect the thermo-chemical coupling effects that dominate in thermo-reactive environments, such as high-temperature oxidation, catalytic reactions, and thermochemical coating processes. In particular, diffusion models applied to metallic and ceramic composites (see Berezovski [12], Gurtin [5]) often treat the reaction and thermal components independently, which oversimplifies the physical process.

Table 1

Symbol	Description	Physical meaning / unit	Range / condition
$\mu(x, t)$	Temperature field	Temperature in the medium (K)	$\mu \geq 0$
$\nu(x, t)$	Concentration of reactive species	Reactant concentration (mol/m ³)	$\nu \geq 0$
χ	Spatial coordinate	Position vector in the domain	$\chi \in \mathbb{R}_+^2$
t	Time variable	Temporal evolution (s)	$t \geq 0$
$\sigma_1(\chi)$	Heat capacity coefficient	Material heat storage ability	$ \chi ^l, l \geq 0$
$\sigma_2(\chi)$	Diffusion coefficient	Thermal or mass diffusivity	$ \chi ^n, n \geq 0$
$\sigma_3(\chi)$	Reaction coefficient	Source or reaction intensity	$ \chi ^k, k \geq 0$
$\varepsilon_1, \varepsilon_2$	Coupling (reaction) parameters	Reaction rate constants	$\varepsilon_i \geq 1$
m_1, m_2	Diffusion nonlinearity exponents	Describe degenerate or fast diffusion	$m_i \geq 1$
k, k_1, k_2	Coupling power indices	Strength of nonlinear interaction	$k_i \geq 1$
l_1, l_2	Gradient exponents	Define nonlinearity in spatial flux	$l_i \geq 1$
p	Diffusion exponent	Order of the p -Laplacian operator	$p \geq 2$
$\mu_0(\chi), \nu_0(\chi)$	Initial conditions	Initial temperature and concentration	Non-negative functions

Recent developments in nonlinear parabolic PDE theory (e.g., DiBenedetto [6], Antontsev [13], Jüngel, [14]) have improved understanding of degenerate and singular diffusion systems, enabling rigorous treatment of variable-coefficient models. Building on these foundations, studies such as Rohde [3] and Huang [15] introduced thermo-chemical coupling terms, but their analyses primarily focused on idealized geometries or one-dimensional configurations.

In contrast, the present work extends these approaches by introducing a two-dimensional nonlinear coupled system with temperature-dependent coefficients $\sigma_1(\chi)$, $\sigma_2(\chi)$, $\sigma_3(\chi)$ and mutual reaction interactions ε_1 , ε_2 . Moreover, while earlier models rely on explicit or semi-implicit numerical schemes, our implementation of an ADI-Peaceman-Rachford method [16] with Thomas tridiagonal optimization achieves higher numerical stability and convergence, even for strongly nonlinear parameter regimes.

From a regional perspective, such models are of particular relevance to Uzbekistan's thermo-chemical engineering sector, where high-temperature alloy diffusion, silicon-based thermal coatings, and reactive gas-solid transformations (e.g., in the Navoi and Olmaliq metallurgical industries) exhibit precisely the type of coupled thermal-chemical dynamics analyzed here. This paper, therefore, bridges the gap between theoretical PDE analysis and applied thermal process modeling, providing both mathematical rigor and technological applicability.

Paper Organization: The remainder of this paper is organized as follows.

Section 1 (Introduction) presents the motivation for studying nonlinear thermo-chemical diffusion systems and outlines the relevance of such models in modern materials science and thermo-reactive process engineering. It also introduces the general structure of the coupled diffusion-reaction equations and highlights the fundamental challenges associated with nonlinear parameter coupling and numerical stability.

Section 2 (Mathematical Formulation of the Thermo-Chemical Diffusion Model) defines the governing equations of the system, namely the coupled parabolic problem for the scalar functions $\mu(x, t)$ and $\nu(x, t)$, together with the associated initial and homogeneous Neumann boundary conditions. The section also specifies the physical meaning of the nonlinear coefficients $\sigma_1(\chi)$, $\sigma_2(\chi)$, $\sigma_3(\chi)$ and interaction terms $\varepsilon_1, \varepsilon_2$, and formulates the well-posedness conditions for the problem.

Section 3 (Numerical Methods for the Nonlinear Diffusion-Reaction System) describes the numerical implementation of the model using an Alternating Direction Implicit (ADI) scheme of the Peaceman-Rachford type, combined with a Thomas tridiagonal solver to efficiently handle the resulting linear systems. Discretization details, stability considerations, and convergence criteria are discussed in relation to both temporal and spatial resolutions.

Section 3.1 (Error Analysis and Stability Assessment of the Computational Scheme) presents the quantitative validation of the proposed numerical algorithm. Error convergence studies, time-step sensitivity tests, and numerical stability analyses are carried out to evaluate the accuracy and robustness of the method. In addition, the section investigates the boundary layer behavior and compares the results with existing benchmark models from recent literature.

Section 4 (Conclusions) summarizes the principal outcomes of the research, emphasizing the successful integration of nonlinear thermo-chemical coupling within a stable computational framework. It also outlines potential directions for future work, including model generalization to three-dimensional geometries and coupling with experimental datasets from real thermo-reactive systems.

Finally, the Acknowledgements section recognizes institutional and funding contributions that supported the study, and the References section provides a comprehensive list of the scientific works that informed and inspired this research.

2. Mathematical formulation of the thermo-chemical diffusion model

In this section, we formulate a generalized nonlinear coupled system that describes the interaction between two evolving fields - the temperature distribution $\mu(x, t)$ and the concentration of reactive species $\nu(x, t)$ - within a spatially heterogeneous medium. Such coupled diffusion-reaction processes appear in a variety of applied problems, including energy transport in composite materials, catalytic reaction systems, and thermal-chemical diffusion phenomena in porous or reactive structures. The governing equations are expressed in a nonlinear form involving several parameters that reflect material properties, diffusion exponents, and coupling strengths. Since the physical interpretation and scaling of these coefficients may vary depending on the specific application, in this work we consider them as generalized parameters whose analytical influence will be studied later.

The complete list of symbols and parameters used in the formulation is summarized in Table 1.

Based on the parameters listed in Table 1, the coupled nonlinear diffusion-reaction system governing the temperature $\mu(x, t)$ and concentration $\nu(x, t)$ fields is formulated as follows:

$$\begin{cases} \sigma_1(\chi) \mu_t = \nabla \left(\sigma_2(\chi) \mu^{m_1-1} |\nabla \chi^k|^{p-2} \nabla \chi^{l_1} \right) + \sigma_3(\chi) \varepsilon_1 \nu^{k_1}, \\ \sigma_1(\chi) \nu_t = \nabla \left(\sigma_2(\chi) \nu^{m_2-1} |\nabla \chi^k|^{p-2} \nabla \chi^{l_2} \right) + \sigma_3(\chi) \varepsilon_2 \mu^{k_2}. \end{cases} \quad (1)$$

The system (1) is supplemented by the following initial conditions:

$$\begin{cases} \mu(0, \chi) = \mu_0(\chi) \geq 0, \\ \nu(0, \chi) = \nu_0(\chi) \geq 0, \quad \chi \in \Omega \subset \mathbb{R}_+^2, \end{cases} \quad (2)$$

and the homogeneous Neumann boundary conditions:

$$\begin{cases} \frac{\partial \mu}{\partial n} = 0, \\ \frac{\partial \nu}{\partial n} = 0, \end{cases} \quad \text{on } \partial\Omega \times (0, T), \quad (3)$$

where n denotes the outward unit normal vector to the boundary $\partial\Omega$.

The coupled nonlinear thermo-chemical diffusion problem described by problem (1)-(3) provides a generalized mathematical framework for a wide class of transport-reaction processes occurring in heterogeneous

and energy-active media [17]. Depending on the physical interpretation of the dependent variables and the choice of coefficients, the model can be applied to several key engineering and scientific systems as follows:

1. Composite and Functionally Graded Materials

In layered or porous composites, $\mu(x, t)$ represents the temperature distribution, while $v(x, t)$ denotes the concentration of a reacting or diffusing substance. The nonlinear diffusion term models the variation of effective thermal conductivity and diffusivity with respect to the material coordinate x , thus capturing anisotropic or temperature-dependent heat transfer effects.

2. Catalytic and Reactive Porous Structures

In catalytic membranes or reactive surfaces, the first equation of problem (1)–(3) governs heat generation and dissipation, whereas the second one represents the diffusion of chemical species involved in exothermic or endothermic reactions. The coupling parameters ε_1 and ε_2 quantify the thermo-chemical feedback between heat and species concentration, which can lead to thermal runaway, self-regulation, or stationary reaction zones.

3. Electrochemical and Energy-Storage Systems

For solid-oxide fuel cells, lithium-ion batteries, or thermally activated sensors, $\mu(x, t)$ corresponds to local thermal energy or temperature, and $v(x, t)$ represents ionic or charge concentration. The nonlinear coupling between these quantities models heat generation due to electrochemical reactions and charge transport under non-isothermal conditions.

4. Environmental and Geophysical Processes

The same problem can describe coupled temperature-moisture transport in soils or pollutant-heat diffusion in groundwater, where x indicates spatial heterogeneity and $\sigma_i(x)$ represent the local variability of the medium's properties.

In summary, problem (1)–(3) can be interpreted as a unified framework for multi-physics thermo-chemical diffusion phenomena, suitable for diverse applications in materials science, environmental modeling, and energy systems engineering. The analytical and numerical investigation of such a problem allows for studying the influence of nonlinear coupling parameters on the stability, propagation, and pattern formation of thermo-reactive fields.

The study of thermo-chemical and coupled diffusion phenomena has attracted growing attention in recent years, particularly due to its relevance in materials science, electrochemical devices, and environmental processes. Various authors have developed and analyzed coupled parabolic systems describing the interaction between thermal, chemical, and mass transport effects. The present subsection provides an overview of several representative works that have significantly contributed to the theoretical and numerical development of such models.

Carrillo [18] proposed a reaction-transport coupled system for analyzing self-excited thermal waves in reactive media, formulated as nonlinear parabolic PDEs linking temperature and concentration through feedback kinetics. Their work, published in *Nature Communications*, demonstrated how data-driven learning can uncover the underlying reaction-transport mechanisms, offering novel insights for the modeling of wildfires and thermal runaway in battery systems. Similarly, Hui [19] developed a coupled electro-thermal reduced-order model describing the interaction between electrochemical potentials and temperature fields in lithium-ion batteries. Their findings, reported in *Applied Energy*, confirmed that reduced-order PDE-ODE hybrid models can efficiently reproduce real-time thermo-electrochemical responses, enabling safer and more stable operation.

Complementary to these contributions, Zhang [20] reviewed recent advances in thermo-electrochemical modeling for thermal runaway prediction, emphasizing the challenges of multi-scale parameter identification. This review consolidated models ranging from macroscopic energy-balance PDEs to microscopic kinetic

reaction systems, identifying critical parameters affecting stability and safety thresholds. Gurtin [5] extended this modeling framework by introducing a fully coupled thermo-electro-chemo-mechanical system, formulated as a set of strongly nonlinear PDEs coupling thermal conduction, ionic diffusion, and mechanical deformation. Their results established a comprehensive computational framework for structural batteries, capturing the complex multiphysics coupling among temperature, concentration, and stress fields.

Huang [15] focused on coupled heat and mass transfer models arising in environmental and civil engineering contexts. They analyzed strongly coupled parabolic systems and introduced numerically stable schemes for their discretization, demonstrating accuracy and robustness under realistic boundary conditions. In parallel, Jüngel [16] developed a boundedness-by-entropy numerical framework for general cross-diffusion systems of the Shigesada-Kawasaki-Teramoto (SKT) type, ensuring discrete entropy dissipation and uniform L^∞ bounds. This approach provides a mathematically rigorous foundation for simulating nonlinear diffusion in multi-component mixtures.

The theoretical background for such models is further enriched by Giachetti [21], who proved the global existence and regularity of strong solutions for SKT-type systems under structural conditions guaranteeing positivity and entropy balance. Their contribution, published in *De Gruyter's analysis series*, is fundamental for establishing well-posedness in thermo-chemical diffusion problems with cross-coupling. Similarly, Smith et al. [2] analyzed degenerate diffusion systems driven by the p -Laplacian operator, revealing refined asymptotic behaviors and scaling limits for nonlinear parabolic equations of the form

$$\partial_t u = \nabla \cdot (|\nabla u|^{p-2} \nabla u)$$

which serve as limiting cases for energy-dissipative diffusion in thermally reactive media.

On the applied side, Fan [22] investigated heat and moisture transport in cement-based materials, modeled by coupled PDE systems capturing energy and mass conservation in porous media. Their work, published in *Energies (MDPI)*, presented experimental validation and parameter sensitivity analysis, showing that accurate coupling between heat and moisture diffusion is essential for predicting material durability and performance. Finally, Owolabi [23] combined machine learning and PDE-constrained modeling to identify hidden reaction-transport couplings from experimental thermal wave data, demonstrating that hybrid data-theory approaches can enhance predictive accuracy in complex thermo-chemical environments.

Collectively, these studies provide a strong theoretical and computational foundation for analyzing thermo-chemical diffusion models. They emphasize the importance of cross-coupling effects, entropy-based stability, and multi-physics interactions - all of which motivate the present work's focus on developing and analyzing a consistent mathematical formulation for coupled heat-mass transfer processes in reactive domains.

To analyze the nonlinear thermo-chemical diffusion problem (1)–(3), we employ the self-similar (automodel) transformation method, which has proven to be a powerful analytical tool for nonlinear parabolic systems. The essential idea of this approach is to reduce the system of partial differential equations to a system of ordinary differential equations by introducing a dimensionless similarity variable that captures the intrinsic scaling of the process.

This transformation allows one to identify invariant relationships between temporal and spatial variables, revealing the fundamental self-organization and propagation mechanisms of the coupled diffusion process. The self-similar framework is especially suitable for problems exhibiting nonlinear diffusion, spatial heterogeneity, or coupled transport effects, as it provides compact analytical

descriptions of asymptotic or intermediate-time behavior [24].

In many physical contexts—such as heat and mass transfer in porous or reactive media, moisture propagation in building materials, or combined chemical-thermal processes—the self-similar method enables the determination of scaling exponents, front evolution laws, and energy-decay characteristics without requiring a full numerical solution. Hence, in the present work, the self-similar technique is adopted as the main analytical approach to investigate the qualitative and asymptotic properties of the thermo-chemical diffusion model.

In order to obtain a self-similar form of the problem (1)–(3), we seek the solution in the automodel form, assuming that the dependent variables $\mu(\ell, \chi)$ and $v(\ell, \chi)$ admit the following similarity transformations:

$$\begin{cases} \mu(\ell, \chi) = (\mathbb{R} + \ell)^{-\gamma_1} \varpi(\lambda(\ell), \chi), \\ v(\ell, \chi) = (\mathbb{R} + \ell)^{-\gamma_2} \theta(\lambda(\ell), \chi), \end{cases} \quad (4)$$

where $\gamma_i = \frac{k_i+1}{k_i k_{3-i}-1}$, $i = 1, 2$.

In the representation (4), the functions ϖ and θ denote the self-similar profile functions (or similarity amplitudes) associated with the main unknowns μ and v . Their mathematical and physical interpretation can be summarized as follows:

- **Definition and role.** The functions $\varpi = \varpi(\lambda, \chi)$ and $\theta = \theta(\lambda, \chi)$ are scalar similarity profiles depending on the similarity variable $\lambda = \lambda(\ell)$ and the spatial coordinate χ . For each fixed χ , they describe the evolution of the scaled amplitudes with respect to λ , while for each fixed λ , they represent the spatial modulation of the diffusive field over the domain Ω .
- **Regularity and positivity.** Throughout the analysis, we assume that

$$\varpi(\cdot, \chi), \theta(\cdot, \chi) \in C^1((0, A]; W^{1,p}(\Omega)) \cap C([0, A]; L^2(\Omega)),$$

homogeneous profiles. The existence, regularity, and asymptotic behavior of ϖ and θ are then analyzed through this reduced similarity system.

- **Physical interpretation.** The profiles ϖ and θ capture the invariant shapes of the temperature and concentration distributions after removing the global algebraic decay expressed by $(\mathbb{R} + \ell)^{-\gamma_i}$. Their large- λ behavior determines the propagation rates of diffusive fronts and the asymptotic decay laws of the original variables μ and v .

It is crucial to observe that the sign of each similarity exponent γ_i in the form (4) serves as a decisive indicator of the long-time behaviour of the solution. In fact, when $\gamma_i < 0$, the algebraic factor $(\mathbb{R} + \ell)^{-\gamma_i}$ grows in time (i.e., $(\mathbb{R} + \ell)^{-\gamma_i} \rightarrow \infty$ as $\ell \rightarrow \infty$), and this growth often indicates that the corresponding field (such as temperature or concentration) becomes unbounded, leading to a finite-time blow-up in the full problem. Conversely, if $\gamma_i > 0$, then $(\mathbb{R} + \ell)^{-\gamma_i} \rightarrow 0$ as $\ell \rightarrow \infty$, which implies algebraic decay of the leading amplitude and therefore favours the global existence (i.e., no blow-up) of the solution.

This dichotomy between the sign of the scaling exponent and the qualitative behaviour of the solution has been rigorously analysed in the context of nonlinear diffusion and weighted reaction-diffusion equations. For example, Kok [25] established global non-blow-up regimes for singular potentials with negative weight exponent $\sigma < 0$, whereas their later work [26] demonstrated finite-time blow-up for supercritical diffusion cases with sufficiently large positive σ . Such results confirm that the structure of the scaling exponents γ_i in the present problem directly determines whether the evolution exhibits self-similar decay or blow-up phenomena.

Hence, by substituting expression (4) into the system of Eq. (1), the following transformed form is obtained:

$$\begin{cases} \frac{\partial \varpi}{\partial \lambda} = \nabla \left(\varpi^{m_1-1} |\nabla \varpi^k|^{p-2} \nabla \varpi^{l_1} \right) + \frac{(\gamma_1/\gamma_2)^{\frac{k_1}{k_1-1}}}{(\gamma_1/\gamma_2)^{\frac{1}{k_2-1}} (1 - \gamma_1(m_1 + l_1 + k(p-2) - 2)) \lambda(\ell)} \left(\frac{\varepsilon_1 \sigma_3}{\sigma_1} \varpi^{k_1} + \gamma_1 \varpi \right), \\ \frac{\partial \theta}{\partial \lambda} = \nabla \left(\theta^{m_2-1} |\nabla \theta^k|^{p-2} \nabla \theta^{l_2} \right) + \frac{(\gamma_1/\gamma_2)^{\frac{-k_2}{k_2-1}}}{(\gamma_1/\gamma_2)^{\frac{1}{k_1-1}} (1 - \gamma_2(m_2 + l_2 + k(p-2) - 2)) \lambda(\ell)} \left(\frac{\varepsilon_2 \sigma_3}{\sigma_1} \varpi^{k_2} + \gamma_2 \theta \right), \end{cases} \quad (5)$$

and that they satisfy the natural physical constraints $\varpi \geq 0$, $\theta \geq 0$ almost everywhere in Ω .

- **Boundary and compatibility conditions.** The profiles ϖ and θ inherit the spatial boundary conditions from the original problem (1)–(3). In particular, they satisfy the homogeneous Neumann conditions on $\partial\Omega$, ensuring zero normal flux:

$$\frac{\partial \varpi}{\partial \chi} = \frac{\partial \theta}{\partial \chi} = 0 \quad \text{on } \partial\Omega.$$

The initial (or normalization) conditions for ϖ and θ at the reference similarity level $\lambda = \lambda_0$ are determined by the initial data μ_0, v_0 via the scaling (4).

- **Reduced problem.** Substituting the transformations (4) into the original system (1)–(3) and collecting terms with respect to powers of $(\mathbb{R} + \ell)$ leads to a reduced system of equations for ϖ and θ . These are typically of mixed parabolic-elliptic type in the variables (λ, χ) or reduce to ordinary differential equations in λ for spatially

Expression (5) corresponds to the spatial scaling in the form of $\lambda(\ell)$:

$$\lambda(\ell) = \begin{cases} D_i (\mathbb{R} + \ell)^{d_i}, & \text{if } \tau_i = m_i + l_i + k(p-2) - 2 \neq 0, \\ \mathbb{R} + \ell, & \text{if } \tau_i = 0, \end{cases} \quad (6)$$

where $A_i = (\gamma_i/\gamma_{3-i})^{\frac{k_i}{k_i-1}}$, $D_i = \frac{A_i^{d_i}}{d_i}$, $d_i = d_{3-i} = 1 - \gamma_i \tau_i$, $i = 1, 2$.

Subsequently, a new transformation of the shape is introduced

$$\begin{cases} \varpi(\lambda(\ell), \chi) = \eta(\zeta), \\ \theta(\lambda(\ell), \chi) = \phi(\zeta), \quad \zeta = \frac{|\chi|}{[\lambda(\ell)]^{1/p}}. \end{cases} \quad (7)$$

If the compatibility condition

$$1 - \gamma_1(m_1 + l_1 + k(p-2) - 2) = 1 - \gamma_2(m_2 + l_2 + k(p-2) - 2) \quad (8)$$

holds, then the multiplicative temporal factors appearing in the right-hand sides of the ϖ - and θ -equations coincide. Consequently, the $\lambda(\ell)$ -dependent prefactors can be absorbed by a single similarity vari-

able and the system (5) admits a reduction to an autonomous self-similar form. In other words, under (8) one may choose the similarity scaling so that all explicit dependences on ℓ (or on the combination $(\mathbb{R} + \ell)$)

$$\begin{cases} -\int_0^T \int_{\Omega} \mu \partial_t \phi \, dx \, dt + \int_0^T \int_{\Omega} D_{11}(\mu, v) \nabla \mu \nabla \phi + D_{12}(\mu, v) \nabla v \nabla \phi \, dx \, dt = \int_0^T \int_{\Omega} f(\mu, v) \phi \, dx \, dt, \\ -\int_0^T \int_{\Omega} v \partial_t \psi \, dx \, dt + \int_0^T \int_{\Omega} D_{21}(\mu, v) \nabla \mu \nabla \psi + D_{22}(\mu, v) \nabla v \nabla \psi \, dx \, dt = \int_0^T \int_{\Omega} g(\mu, v) \psi \, dx \, dt, \end{cases} \quad (10)$$

vanish from the reduced equations, and the profiles $\varpi(\lambda, \chi)$ and $\theta(\lambda, \chi)$ satisfy a closed system of similarity equations independent of the outer time-scale.

This compatibility is therefore a necessary algebraic condition for the existence of a true automodel reduction: when (8) is satisfied the transformed problem is self-consistent and amenable to standard similarity analysis (existence, regularity and asymptotics of the profile functions);

if (8) fails, the reduced equations retain an explicit nontrivial $\lambda(\ell)$ -dependence and a genuine self-similar reduction of the form (5) is not possible.

From the obtained results, we derive the following self-similar form of the equations for the functions $\eta(\zeta)$, $\phi(\zeta)$.

$$\begin{cases} \frac{d}{d\zeta} \left(\eta^{m_1-1} \left| \frac{d\eta^k}{d\zeta} \right|^{p-2} \frac{d\eta^l}{d\zeta} \right) + \frac{\zeta}{p} \frac{d\eta}{d\zeta} + \beta_1 \left(\frac{\varepsilon_1 \sigma_3}{\sigma_1} \phi^{k_1} + \gamma_1 \eta \right) = 0, \\ \frac{d}{d\zeta} \left(\phi^{m_2-1} \left| \frac{d\phi^k}{d\zeta} \right|^{p-2} \frac{d\phi^l}{d\zeta} \right) + \frac{\zeta}{p} \frac{d\phi}{d\zeta} + \beta_2 \phi \left(\frac{\varepsilon_2 \sigma_3}{\sigma_1} \eta^{k_2} + \gamma_2 \phi \right) = 0, \end{cases} \quad (9)$$

were

$$\begin{cases} \beta_1 = \frac{(\gamma_1/\gamma_2)^{\frac{k_1}{k_1-1}}}{(\gamma_1/\gamma_2)^{\frac{1}{k_2-1}} (1 - \gamma_1(m_1 + l_1 + k(p-2) - 2))}, \\ \beta_2 = \frac{(\gamma_1/\gamma_2)^{\frac{-k_2}{k_2-1}}}{(\gamma_1/\gamma_2)^{\frac{1}{k_1-1}} (1 - \gamma_2(m_2 + l_2 + k(p-2) - 2))}. \end{cases}$$

The determination of self-similar solutions plays a crucial role in the qualitative analysis of nonlinear diffusion and thermo-chemical processes. Such solutions often describe the asymptotic behavior of the system at large times or near singularities, revealing the dominant mechanisms of heat and mass transfer within the considered physical domain. In particular, the self-similar reduction transforms the original nonlinear PDE system into a set of ordinary differential equations, enabling analytical or semi-analytical investigation of the profile functions.

Furthermore, establishing theorems concerning the existence, uniqueness, and stability of these self-similar solutions is essential for understanding the dynamic evolution of the system. These results allow one to characterize whether the physical process exhibits global regularity, finite-time blow-up, or steady propagation fronts, thereby providing a rigorous mathematical framework for observing and predicting the motion of the thermo-chemical front in heterogeneous media.

Definition 1. Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary and let $T > 0$. A pair of functions

$$(\mu, v) \in L^p(0, T; W^{1,p}(\Omega)) \cap C([0, T]; L^2(\Omega))$$

is called a weak solution of system (1)-(3) if, for all test functions $\phi, \psi \in C_0^\infty(\Omega \times [0, T])$, the following integral equalities hold:

together with the initial conditions $\mu(x, 0) = \mu_0(x)$ and $v(x, 0) = v_0(x)$ in $L^2(\Omega)$.

Theorem 1. Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with C^1 boundary and let $T > 0$. Assume $p \geq 2$ and the model exponents satisfy $m_i, k_i, l_i \geq 1$ for $i = 1, 2$. Suppose the spatial coefficients $\sigma_j(\chi)$ ($j = 1, 2, 3$) are uniformly positive and bounded a.e. in Ω , the reaction terms $R_1(\mu, v) = \sigma_3(\chi) \varepsilon_1 v^{k_1}$ and $R_2(\mu, v) = \sigma_3(\chi) \varepsilon_2 \mu^{k_2}$ are locally Lipschitz and obey a polynomial growth bound below the Sobolev critical exponent in two dimensions, and the initial data satisfy $\mu_0, v_0 \in L^2(\Omega)$ with $\mu_0, v_0 \geq 0$ a.e.

Then there exists at least one weak solution

$$\mu, v \in L^2(0, T; W^{1,p}(\Omega)) \cap C([0, T]; L^2(\Omega))$$

of problem (1)-(3) which satisfies the equations in the usual variational sense and the homogeneous Neumann boundary conditions. Moreover, if the reaction maps are globally Lipschitz, the weak solution is unique.

Proof. The proof is based on the standard Galerkin approximation, compactness arguments and the monotonicity method for nonlinear parabolic systems.

Step 1. Galerkin approximation. Let $\{w_j\}_{j=1}^\infty$ be a complete orthonormal basis in $W^{1,p}(\Omega)$ that is also orthogonal in $L^2(\Omega)$. For each $N \in \mathbb{N}$, we consider the approximate solutions

$$\begin{cases} \mu_N(x, t) = \sum_{j=1}^N a_{jN}(t) w_j(x), \\ v_N(x, t) = \sum_{j=1}^N b_{jN}(t) w_j(x), \end{cases} \quad (11)$$

where the coefficients $a_{jN}(t), b_{jN}(t)$ solve the finite-dimensional system obtained by projecting (1)-(3) onto $\text{span}\{w_1, \dots, w_N\}$. Due to the local Lipschitz continuity of the nonlinearities and the boundedness of coefficients, classical ODE theory guarantees the existence of a local solution $(a_{jN}(t), b_{jN}(t))$ on some time interval $[0, T_N)$.

Step 2. A priori estimates. Multiplying the first equation of the approximate system by μ_N and the second by v_N , integrating over Ω , and using integration by parts with the homogeneous Neumann boundary conditions, we obtain the energy inequality

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} (\|\mu_N\|_{L^2}^2 + \|v_N\|_{L^2}^2) + \int_{\Omega} (|\nabla \mu_N^{k_1}|^p + |\nabla v_N^{k_2}|^p) \, dx \\ \leq C \int_{\Omega} (|\mu_N|^2 + |v_N|^2) \, dx, \end{aligned} \quad (12)$$

where $C > 0$ depends only on the parameters $\sigma_i(\chi), m_i, l_i, k_i$. Applying Gronwall's inequality and noting that $\mu_0, v_0 \in L^2(\Omega)$, we deduce uniform bounds

$$\mu_N, v_N \text{ are bounded in } L^\infty(0, T; L^2(\Omega)) \cap L^p(0, T; W^{1,p}(\Omega)).$$

This implies global existence of the approximate solutions on $[0, T]$.

Step 3. Compactness and limit passage. By the uniform energy

bounds and the Aubin-Lions compactness lemma ([27]), there exist subsequences (not relabeled) such that

$$\mu_N \rightarrow \mu, \quad v_N \rightarrow v \quad \text{strongly in } L^2(0, T; L^2(\Omega)),$$

and weakly in $L^p(0, T; W^{1,p}(\Omega))$. Furthermore, the nonlinear diffusion terms converge weakly in the dual space $L^{p'}(0, T; W^{-1,p'}(\Omega))$ due to the monotonicity of the p -Laplace type operators.

Step 4. Passing to the limit. Using the weak convergence results and the weak lower semicontinuity of convex functionals, we can pass to the limit in the variational formulation of the approximate system. The limit functions (μ, v) satisfy the weak form of the original system (1)–(3) in the sense of Definition 1 (weak solution). The energy inequality also passes to the limit, guaranteeing that the weak solution inherits the same L^2 and $W^{1,p}$ bounds.

Step 5. Uniqueness (conditional). If the nonlinear reaction operators are globally Lipschitz or satisfy a monotonicity condition of the form

$$(R_i(u_1, v_1) - R_i(u_2, v_2))(u_1 - u_2) \geq 0, \quad i = 1, 2, \quad (13)$$

then testing the difference of two weak solutions with $\mu_1 - \mu_2$ and $v_1 - v_2$ and integrating yields a Gronwall-type inequality implying $\mu_1 = \mu_2$, $v_1 = v_2$ a.e. in Ω_T .

Conclusion. Hence, by the Galerkin approximation, uniform a priori estimates, and compactness arguments, we conclude that the system (1)–(3) admits at least one weak solution

$$(\mu, v) \in L^2(0, T; W^{1,p}(\Omega)) \cap C([0, T]; L^2(\Omega)),$$

which is unique under an additional Lipschitz or monotonicity assumption on the nonlinear source terms.

Remark 1. Theorem 1 provides a rigorous foundation for the mathematical modelling of the specific thermo-chemical object under consideration: a heterogeneous reactive medium in which temperature-like and concentration-like fields interact through strongly nonlinear diffusion and reaction couplings. Practically, the theorem guarantees that, under the stated structural hypotheses on the diffusion exponents, spatial coefficients and reaction terms, the model yields a well-posed evolution (existence of a weak solution) and, under additional monotonicity or Lipschitz conditions, uniqueness.

From the modelling viewpoint this result is instrumental in several respects. First, it legitimizes the use of the proposed PDE system as a predictive tool: numerical simulations and parameter identification carried out within the theorem's admissible parameter regime are founded on solutions that exist and depend continuously on the data. Second, the theorem delineates the parameter regimes in which diffusion dominates (leading to global-in-time, controlled evolution) versus regimes where nonlinear source terms may drive the system to loss of control (finite-time blow-up). This dichotomy directly informs experimental design, operational limits, and control strategies by indicating which combinations of material heterogeneity, initial loading, and reaction strength are safe or critical.

As a concrete, locally relevant example, consider industrial brick firing in Uzbekistan. Let μ denote the temperature field inside a porous brick, and let v denote the pore moisture content or a volatile reaction product. During fast heating in a kiln, strong temperature gradients, moisture evaporation and exothermic chemical changes may interact. Theorem 1 implies that if the effective parameters of the brick-kiln system (diffusion nonlinearity, coupling exponents and material heterogeneity captured by $\sigma_j(\chi)$) satisfy the theorem's hypotheses and the resulting similarity exponents γ_i are positive, then the heating process is predicted to remain globally regular: moisture and temperature evolve without singular growth, reducing the risk of catastrophic internal steam pressure or thermal cracking. Conversely, if the same parameters yield $\gamma_i < 0$, the model signals a regime in which uncontrolled temperature or

pressure escalation (mathematically: blow-up) is possible, thereby indicating the need to reduce heating rates, pre-dry the bricks, or enhance kiln ventilation. Thus, Theorem 1 not only secures mathematical well-posedness but also provides actionable criteria for process safety and parameter selection in practical operations typical for Uzbekistan's ceramic and brick industries.

Theorem 2. Let the system (1)–(3) satisfy the regularity conditions assumed in Theorem 1, and suppose that the initial data $\mu_0(\ell, \chi) \geq 0$ and $v_0(\ell, \chi) \geq 0$ for all $(\ell, \chi) \in \Omega \times (0, T]$. If the nonlinear diffusion coefficients $D_{ij}(\mu, v)$ and the reaction terms $f(\mu, v)$, $g(\mu, v)$ are locally Lipschitz continuous and satisfy

$$f(0, 0) \geq 0, \quad g(0, 0) \geq 0,$$

then the weak solution $(\mu(\ell, \chi), v(\ell, \chi))$ of problem remains nonnegative for all $(\ell, \chi) \in \Omega \times (0, T]$. Moreover, there exists a constant $C > 0$, depending only on the initial data and model parameters, such that

$$0 \leq \mu(\ell, \chi) \leq C, \quad 0 \leq v(\ell, \chi) \leq C, \quad \forall (\ell, \chi) \in \Omega \times (0, T].$$

Hence, the thermo-chemical diffusion process described by system preserves positivity and boundedness throughout its evolution.

Proof. The arguments follow classical techniques for nonlinear parabolic systems: testing by sign-parts for positivity (Stampacchia type argument / comparison principle) and energy/Moser iteration for L^∞ bounds (see e.g. [4,11,18,28,29]).

Step 1. Preservation of nonnegativity. Let (μ, v) be a weak solution given by Theorem 1. Define the negative parts

$$\mu^- := \max\{-\mu, 0\}, \quad v^- := \max\{-v, 0\}.$$

These functions belong to $L^2(0, T; W^{1,p}(\Omega))$ and vanish at $t = 0$ by the hypothesis $\mu_0, v_0 \geq 0$ a.e. Test the weak formulation of the μ -equation with $\phi = \mu^-$ and the v -equation with $\psi = v^-$. Using integration by parts and the homogeneous Neumann boundary conditions we obtain (formally)

$$\begin{aligned} & \int \frac{1}{2} \frac{d}{dt} \|\mu^-(t)\|_{L^2(\Omega)}^2 + \int_{\Omega} A_{\mu}(\nabla \mu, \mu, v) \cdot \nabla \mu^- dx = \int_{\Omega} R_1(\mu, v) \mu^- dx, \\ & \int \frac{1}{2} \frac{d}{dt} \|v^-(t)\|_{L^2(\Omega)}^2 + \int_{\Omega} A_v(\nabla v, \mu, v) \cdot \nabla v^- dx = \int_{\Omega} R_2(\mu, v) v^- dx, \end{aligned} \quad (14)$$

where A_{μ}, A_v denote the nonlinear diffusion fluxes. By the structure of the p -Laplacian-type fluxes (coercivity and monotonicity) the gradient terms are nonnegative in the sense

$$\int_{\Omega} A_{\mu}(\nabla \mu, \cdot) \cdot \nabla \mu^- dx \geq 0, \quad \int_{\Omega} A_v(\nabla v, \cdot) \cdot \nabla v^- dx \geq 0. \quad (15)$$

Concerning the reaction terms, under our model assumptions $R_1(\mu, v) = \sigma_3(\chi) \varepsilon_1 v^{k_1} \geq 0$ and $R_2(\mu, v) = \sigma_3(\chi) \varepsilon_2 \mu^{k_2} \geq 0$ whenever $\mu, v \geq 0$. Moreover, by local Lipschitz continuity and the hypothesis $f(0, 0) \geq 0$, $g(0, 0) \geq 0$, one verifies that on the support of μ^- (where $\mu \leq 0$ and $v \geq -v^-$) the terms $R_1 \mu^-$ and $R_2 v^-$ are nonpositive or can be bounded above by zero. Thus, summing the two identities yields

$$\frac{d}{dt} (\|\mu^-(t)\|_{L^2}^2 + \|v^-(t)\|_{L^2}^2) \leq 0. \quad (16)$$

Since $\mu^-(0) = v^-(0) = 0$, it follows that $\mu^-(t) \equiv v^-(t) \equiv 0$ for all $t \in [0, T]$. Hence $\mu, v \geq 0$ a.e. in $\Omega \times (0, T]$. The above sign-preservation argument for coupled parabolic systems is standard; see e.g. [14,20,30–32] and comparison principles in [12].

Step 2. L^∞ -boundedness. We now obtain uniform-in-time L^∞ bounds. From Theorem 1 and the a priori energy estimates one has

$$\mu, v \in L^\infty(0, T; L^2(\Omega)) \cap L^p(0, T; W^{1,p}(\Omega)),$$

with constants depending on the initial data and structural parameters.

To lift these bounds to L^∞ we follow a Moser-iteration (or De Giorgi) scheme adapted to degenerate parabolic equations (cf. [26,33]).

Fix $r \geq 2$ and test the μ -equation with $\phi = \mu^{r-1}$ (a standard admissible test after approximation). Integrating over Ω and using monotonicity/coercivity of the diffusion operator yields an inequality of the form

$$\frac{d}{dt} \|\mu\|_{L^r}^r + C_1 \int_{\Omega} |\nabla \mu^{\alpha_r}|^p dx \leq C_2 (\|\mu\|_{L^r}^r + \|v\|_{L^{\beta_r}}^{\beta_r} + 1), \quad (17)$$

for suitable exponents $\alpha_r, \beta_r > 0$ and constants $C_1, C_2 > 0$ depending on the model parameters. A similar inequality holds for v . Using Sobolev embeddings and interpolation inequalities one can derive a recursive inequality for the sequence of norms $\|\mu\|_{L^{r_n}}$ with $r_{n+1} = \kappa r_n$ for some $\kappa > 1$. Iterating and letting $n \rightarrow \infty$ yields a finite upper bound for $\|\mu\|_{L^\infty(\Omega \times (0, T))}$, provided the initial L^2 -mass is finite and the growth of source terms is below the Sobolev critical exponent. The precise implementation of this bootstrap is classical; see [7,10,34] and the Moser iteration technique in [35]. Vázquez gives a detailed treatment for porous-medium type and related degenerate equations [14].

Therefore there exists $C > 0$, depending only on the data and structural constants, such that

$$0 \leq \mu(\ell, \chi), \quad v(\ell, \chi) \leq C \quad \text{a.e. in } \Omega \times (0, T].$$

Remark 2. The results established in Theorem 2 not only provide the theoretical guarantee of solution existence and stability, but also offer practical insights into the real physical and engineering systems under consideration. Below, two representative examples are discussed to illustrate the applicability of the obtained results to both local (Uzbekistan-specific) and global contexts.

(1) Local example - Industrial brick kiln process in Uzbekistan.

In modern ceramic and construction industries, the high-temperature firing of porous clay materials involves rapid heating, moisture evaporation, and exothermic reactions occurring simultaneously within a confined medium. Theorem 2 ensures that, under the prescribed diffusion and reaction parameters, the temperature and moisture distributions inside the material evolve in a controlled manner, avoiding internal pressure accumulation and thermal cracking. If, however, the characteristic exponents $\gamma_i < 0$, the system tends toward a blow-up regime, indicating a risk of structural failure or uncontrolled heat propagation. From a practical viewpoint, the measurable parameters include the initial moisture content v_0 , the initial temperature profile μ_0 , the heating rate ($^\circ\text{C}/\text{min}$), the effective thermal conductivity and porosity (through σ -functions), and the reaction density coefficients. Hence, the theorem provides a theoretical basis for defining safety thresholds such as limiting the heating gradient, implementing pre-drying stages, and controlling ventilation and pressure thereby preventing instability during the sintering process.

(2) Global example - Thermal stability in lithium-ion battery systems.

In large-scale lithium-ion battery modules used in electric vehicles (EV) and energy storage systems (BESS), complex electrochemical and thermal interactions may lead to localized heat accumulation, gas generation, and potential thermal runaway. Theorem 2 provides a sufficient condition for global existence of the solution, ensuring that the temperature and concentration fields remain bounded when diffusion and reaction exponents satisfy the stability criterion ($\gamma_i > 0$). Conversely, when $\gamma_i < 0$, the model predicts finite-time blow-up, which physically corresponds to an uncontrolled temperature increase leading to safety hazards. In practice, the relevant parameters to monitor are the internal cell temperature (via

thermistors), internal pressure and gas release rate, current intensity, electrode thermal conductivity, and reaction kinetics coefficients. Thus, the theorem serves as a mathematical tool for designing control strategies such as thermal management systems, emergency shutdown protocols, and early-warning monitoring-based on the proximity of system parameters to critical values of γ_i .

Theorem 3. Let the coupled thermo-chemical diffusion problem (1)-(3) be considered in a bounded, sufficiently smooth domain $\Omega \subset \mathbb{R}^n$ with initial data $u_0(x), v_0(x) \geq 0$, and let the diffusion coefficients $D_{ij}(u, v)$ and reaction functions $f(u, v), g(u, v)$ be locally Lipschitz continuous and satisfy the polynomial growth condition. If the parameters $\gamma_i > 0$ for $i = 1, 2$, and the entropy functional

$$E(t) = \int_{\Omega} (u^{r_1} + v^{r_2}) dx \quad (18)$$

is non-increasing in time, then the weak solution (u, v) of the problem (1)-(3) exists globally in time and remains uniformly bounded in $L^\infty(\Omega \times (0, T))$ for any finite $T > 0$.

Conversely, if $\gamma_i < 0$ for some $i \in 1, 2$, then the entropy functional $E(t)$ becomes unbounded as $t \rightarrow T^*$ for some finite T^* , indicating a finite-time blow-up in the corresponding component.

Proof. The proof splits into the (A) global boundedness part and the (B) blow-up part. The main ideas combine entropy/energy estimates, ODE comparison and classical bootstrapping arguments; the approach closely follows entropy methods in cross-diffusion literature [36] and nonlinear diffusion theory [21,37].

(A) **Global boundedness for $\gamma_i > 0$.** Under the structural hypotheses on the diffusion fluxes and reaction terms (local Lipschitz, polynomial growth below Sobolev criticality), one may test the weak formulation with suitable powers of u and v to derive an evolution law for $E(t)$. A standard computation (multiply the u -equation by $\gamma_1 u^{r_1-1}$ and the v -equation by $\gamma_2 v^{r_2-1}$, integrate over Ω and sum) yields an entropy balance of the form

$$\frac{d}{dt} E(t) + \underbrace{\int_{\Omega} (a_1 |\nabla u^{\alpha_1}|^p + a_2 |\nabla v^{\alpha_2}|^p) dx}_{\mathcal{D}(t)} \leq C \int_{\Omega} (u^{r_1} + v^{r_2}) dx, \quad (19)$$

where $\alpha_i > 0$ and $a_i > 0$ depend on k_i, m_i, l_i, p and the structural coefficients; the right-hand side arises from lower-order reaction/source contributions and can be absorbed when $\gamma_i > 0$ by choosing the entropy exponents consistently with the diffusion scaling (see e.g. [25]). Thus we obtain

$$\frac{d}{dt} E(t) + C_1 \mathcal{D}(t) \leq C_4 E(t), \quad (20)$$

for some constants $C_1, C_4 > 0$. Gronwall's lemma combined with the dissipation term implies $E(t)$ cannot grow faster than exponentially; more precisely, one obtains an a priori bound for $E(t)$ on any finite time interval.

Next, the dissipation $\mathcal{D}(t)$ controls Sobolev norms of u^{α_1} and v^{α_2} . By Sobolev embedding and interpolation inequalities one obtains higher integrability of u and v (cf. [31,38]). Applying a standard Moser-iteration or De Giorgi scheme (adapted to the degenerate p -Laplacian structure) upgrades L^q -bounds to L^∞ -bounds on finite time intervals; this step is classical in the theory of degenerate parabolic equations (see [39–41]). Jüngel's boundedness-by-entropy framework gives an elegant route to obtain uniform L^∞ bounds under an entropy dissipation inequality of the above type; see [31] for precise hypotheses and implementation. Therefore, (u, v) remains uniformly bounded on

$\Omega \times (0, T)$ for any finite T , and global existence follows from standard continuation arguments (solutions extend while norms remain finite).

(B) **Finite-time blow-up for $\gamma_i < 0$.** If some $\gamma_i < 0$, the same entropy computation produces terms with the opposite sign: the algebraic prefactor corresponding to $(\mathcal{R} + \mathcal{E})^{-\gamma_i}$ grows in time and feeds energy into the entropy. Under the model assumptions this leads to an inequality of the form

$$\frac{d}{dt}E(t) \geq C_2 \int_{\Omega} \Phi(u, v) dx - C_3, \quad (21)$$

where $\Phi(u, v)$ is a positive functional which (by Holder/interpolation) dominates a power of $E(t)$, i.e.

$$\int_{\Omega} \Phi(u, v) dx \geq C_5 E(t)^{1+\delta} \quad (22)$$

for some $\delta > 0$ and $C_5 > 0$ depending on the exponents and domain (this reduction is standard, see e.g. Fujita/Levine type arguments for semi-linear problems [42,43] and weighted diffusion blow-up analyses [44]). Hence

$$\frac{d}{dt}E(t) \geq \tilde{C} E(t)^{1+\delta} - C_3. \quad (23)$$

By ODE comparison (cf. [2,14,45,46]), if $E(0)$ is sufficiently large (or if the positive term dominates C_3), the solution of the scalar ODE

$$y'(t) = \tilde{C} y(t)^{1+\delta} - C_3, \quad y(0) = E(0),$$

blows up in finite time. Therefore $E(t)$, and consequently at least one of the components u or v , becomes unbounded in finite time. This concludes the blow-up part.

Remark 3. The implications of Theorem 3 are highly relevant for the thermochemical diffusion processes occurring in heterogeneous porous and composite structures. In particular, the established global-blow-up dichotomy provides a rigorous mathematical framework for understanding whether a coupled heat-mass transfer process in a given medium stabilizes or undergoes thermal runaway.

From the standpoint of the considered object - for instance, the thermochemical diffusion within the structural materials of solar concentrator towers, ceramic composites, or high-temperature heat storage modules - the parameters γ_i encapsulate the balance between energy absorption, diffusivity, and reaction kinetics. When $\gamma_i > 0$, the diffusion dominates the nonlinear source effects, corresponding physically to stable temperature and concentration profiles that evolve toward steady or self-similar regimes. This reflects operational stability in heat-storage composites and prevents overheating or degradation, as seen in thermal energy storage systems operating in arid regions.

Conversely, if $\gamma_i < 0$, the model predicts a blow-up regime, indicating rapid temperature escalation or concentration spikes due to insufficient dissipation. Such conditions correspond to material failure or uncontrolled thermal propagation. This theoretical prediction aligns with experimental observations in highly nonuniform porous ceramics and phase-change materials, where local overheating can lead to micro-structural damage or ignition phenomena.

Example. (Uzbekistan context). In the context of Uzbekistan, Theorem 3 can be directly interpreted for thermochemical diffusion within solar salt storage tanks used in hybrid solar-thermal power plants located in

Navoi and Bukhara regions. During continuous solar charging, if the internal energy release and diffusion exponents correspond to $\gamma_i > 0$, the thermal field remains globally bounded, ensuring stable storage and gradual heat release. However, in cases where $\gamma_i < 0$ (e.g., due to excessive local heat generation or poor material conductivity), the temperature field may experience finite-time blow-up, leading to salt stratification or local melting. These outcomes are in qualitative agreement with computational results reported for nonlinear heat transfer in inhomogeneous media by Matyakubov [47] and with recent simulation studies on molten salt instabilities in concentrated solar power systems.

Global relevance. Similar phenomena are observed worldwide in energy and environmental systems where strong thermochemical coupling occurs. For instance:

- In geothermal reservoirs and deep rock formations (Iceland, Japan), localized blow-up corresponds to rapid pressure and heat build-up leading to micro-seismic events.
- In lithium-ion battery packs and fuel cells, finite-time blow-up analogs describe thermal runaway and electrolyte degradation.
- In carbon capture and storage (CCS) projects, bounded regimes correspond to stable diffusion of injected CO_2 in porous strata, while negative γ_i scenarios may lead to unstable plume migration or channeling.

Thus, Theorem 3 not only provides the mathematical basis for the stability or blow-up of the considered system, but also serves as a diagnostic tool for evaluating the safety and operational limits of thermochemical processes in engineering and environmental applications. In this sense, it bridges rigorous PDE theory with the design and optimization of modern energy technologies.

3. Numerical methods for the nonlinear diffusion-reaction system

To investigate the spatiotemporal dynamics governed by the nonlinear coupled system (1)-(3), we construct stable and efficient finite difference schemes based on alternating direction implicit (ADI) strategies and variable-direction splitting methods. Considering the strong nonlinearities in diffusion and reaction terms, as well as heterogeneous coefficients, the Peaceman-Rachford approach is adopted to decouple the multi-dimensional operators while preserving the essential nonlinear structure. Each semi-step yields a tridiagonal system that can be efficiently solved via the Thomas algorithm ([23,48]).

Let the spatial domain $\Omega = [0, L_x] \times [0, L_y]$ be discretized into N_x and N_y uniform nodes with spatial steps Δx and Δy , respectively, and the temporal domain $[0, T]$ be divided into N_t time layers with step Δt . The grid points are defined as

$$x_i = i\Delta x, \quad y_j = j\Delta y, \quad t_n = n\Delta t,$$

where $i = 0, 1, \dots, N_x$, $j = 0, 1, \dots, N_y$, and $n = 0, 1, \dots, N_t$.

We approximate the solutions as

$$\begin{cases} u_{ij}^n \approx u(x_i, y_j, t_n), \\ v_{ij}^n \approx v(x_i, y_j, t_n). \end{cases}$$

To preserve conservativity we compute face-centered diffusion weights. Define at time level n the auxiliary discrete gradients

$$\nabla_h (\mu^{k_1})_{i+\frac{1}{2},j}^n := \left(\frac{(\mu_{i+1,j}^n)^{k_1} - (\mu_{i,j}^n)^{k_1}}{h_x}, \frac{(\mu_{i+1,j+1}^n)^{k_1} - (\mu_{i+1,j-1}^n)^{k_1} + (\mu_{i,j+1}^n)^{k_1} - (\mu_{i,j-1}^n)^{k_1}}{4h_y} \right),$$

and similarly for $\nabla_h(v^{k_2})$. Then set

$$\begin{cases} \mu_{i+\frac{1}{2}j}^{(D)} := \sigma_2 \left(x_{i+\frac{1}{2}j}, y_j \right) \left(\frac{\mu_{i+1j}^n + \mu_{ij}^n}{2} \right)^{m_1-1} |\nabla_h(\mu^k)_{i+\frac{1}{2}j}|^{p-2}, \\ \nu_{i+\frac{1}{2}j}^{(D)} := \sigma_2 \left(x_{i+\frac{1}{2}j}, y_j \right) \left(\frac{v_{i+1j}^n + v_{ij}^n}{2} \right)^{m_2-1} |\nabla_h(v^k)_{i+\frac{1}{2}j}|^{p-2}. \end{cases} \quad (24)$$

Analogous definitions hold for $i, j + \frac{1}{2}$ faces.

We use two fractional substeps per time step (Peaceman-Rachford).

First half-step: implicit in x , explicit in y .

Half-step 1 (implicit in x).

For each fixed j solve for $\mu_{ij}^{n+1/2}$ and $\nu_{ij}^{n+1/2}$ from

$$\begin{aligned} \sigma_{1,ij}^n \frac{\mu_{ij}^{n+1/2} - \mu_{ij}^n}{\tau/2} &= \frac{1}{h_x} \left(\mu_{i+\frac{1}{2}j}^{(D)} \frac{\mu_{i+1j}^{n+1/2} - \mu_{ij}^{n+1/2}}{h_x} - \mu_{i-\frac{1}{2}j}^{(D)} \frac{\mu_{ij}^{n+1/2} - \mu_{i-1j}^{n+1/2}}{h_x} \right) \\ &+ \frac{1}{h_y} \left(\mu_{ij+\frac{1}{2}}^{(D)} \frac{\mu_{ij+1}^n - \mu_{ij}^n}{h_y} - \mu_{ij-\frac{1}{2}}^{(D)} \frac{\mu_{ij}^n - \mu_{ij-1}^n}{h_y} \right) + \sigma_{3,ij}^n \varepsilon_1 (v_{ij}^n)^{k_1}, \end{aligned} \quad (25)$$

and similarly

$$\mathbf{f}_i = \left[\sigma_{1,ij}^n \mu_{ij}^n + \frac{\tau}{2} \left(\frac{\mu_{ij+\frac{1}{2}}^{(D)} \left(\mu_{ij+1}^n - \mu_{ij}^n \right) - \mu_{ij-\frac{1}{2}}^{(D)} \left(\mu_{ij}^n - \mu_{ij-1}^n \right)}{h_y^2} \right) + \frac{\tau}{2} \sigma_{3,ij}^n \varepsilon_1 (v_{ij}^n)^{k_1} \right].$$

(same for v component)

$$\begin{aligned} \sigma_{1,ij}^n \frac{v_{ij}^{n+1/2} - v_{ij}^n}{\tau/2} &= \frac{1}{h_x} \left(\nu_{i+\frac{1}{2}j}^{(D)} \frac{v_{i+1j}^{n+1/2} - v_{ij}^{n+1/2}}{h_x} - \nu_{i-\frac{1}{2}j}^{(D)} \frac{v_{ij}^{n+1/2} - v_{i-1j}^{n+1/2}}{h_x} \right) \\ &+ \frac{1}{h_y} \left(\nu_{ij+\frac{1}{2}}^{(D)} \frac{v_{ij+1}^n - v_{ij}^n}{h_y} - \nu_{ij-\frac{1}{2}}^{(D)} \frac{v_{ij}^n - v_{ij-1}^n}{h_y} \right) + \sigma_{3,ij}^n \varepsilon_2 (\mu_{ij}^n)^{k_2}. \end{aligned} \quad (26)$$

For each fixed j the two formulas (25),(26) are tridiagonal in the index i and can be written in block form

$$A_i \mathbf{w}_{i-1} + B_i \mathbf{w}_i + C_i \mathbf{w}_{i+1} = \mathbf{f}_i, \quad \mathbf{w}_i = \begin{bmatrix} \mu_{ij}^{n+1/2} \\ \nu_{ij}^{n+1/2} \end{bmatrix},$$

where the diagonal blocks B_i and off-diagonals depend on $\mu_{i\pm 1/2j}^{(D)}, \nu_{i\pm 1/2j}^{(D)}$ and $\sigma_{1,ij}^n$ (explicit formulae given below).

Half-step 2 (implicit in y).

Compute face coefficients at level $n+1/2$ (re-evaluate gradients with $\mu^{n+1/2}, v^{n+1/2}$), then for each fixed i solve

$$\begin{aligned} \sigma_{1,ij}^{n+1/2} \frac{\mu_{ij}^{n+1} - \mu_{ij}^{n+1/2}}{\tau/2} &= \frac{1}{h_x} \left(\tilde{\mu}_{i+\frac{1}{2}j}^{(D)} \frac{\mu_{i+1j}^{n+1/2} - \mu_{ij}^{n+1/2}}{h_x} - \tilde{\mu}_{i-\frac{1}{2}j}^{(D)} \frac{\mu_{ij}^{n+1/2} - \mu_{i-1j}^{n+1/2}}{h_x} \right) \\ &+ \frac{1}{h_y} \left(\tilde{\mu}_{ij+\frac{1}{2}}^{(D)} \frac{\mu_{ij+1}^{n+1} - \mu_{ij}^{n+1}}{h_y} - \tilde{\mu}_{ij-\frac{1}{2}}^{(D)} \frac{\mu_{ij}^{n+1} - \mu_{ij-1}^{n+1}}{h_y} \right) \\ &+ \sigma_{3,ij}^{n+1/2} \varepsilon_1 (v_{ij}^{n+1/2})^{k_1}, \end{aligned} \quad (27)$$

and analogously for v_{ij}^{n+1} . For each fixed i these equations are tridiagonal in index j .

Writing (25) (for fixed j) in matrix form yields for the μ -component coefficients (simplified notation):

$$\begin{aligned} A_i &= -\frac{\tau}{2h_x^2} \begin{bmatrix} \mu_{i-\frac{1}{2}j}^{(D)} & 0 \\ 0 & \nu_{i-\frac{1}{2}j}^{(D)} \end{bmatrix}, \quad C_i = -\frac{\tau}{2h_x^2} \begin{bmatrix} \mu_{i+\frac{1}{2}j}^{(D)} & 0 \\ 0 & \nu_{i+\frac{1}{2}j}^{(D)} \end{bmatrix}, \\ B_i &= \begin{bmatrix} \sigma_{1,ij}^n + \frac{\tau}{2h_x^2} \left(\mu_{i-\frac{1}{2}j}^{(D)} + \mu_{i+\frac{1}{2}j}^{(D)} \right) & 0 \\ 0 & \sigma_{1,ij}^n + \frac{\tau}{2h_x^2} \left(\nu_{i-\frac{1}{2}j}^{(D)} + \nu_{i+\frac{1}{2}j}^{(D)} \right) \end{bmatrix}, \end{aligned}$$

and the right-hand side vector $\mathbf{f}_i$ collects explicit y -flux terms and source couplings:

The block Thomas algorithm is applied to the sequence $\{A_i, B_i, C_i, \mathbf{f}_i\}_{i=1}^{N_x-1}$ for each fixed j , then similarly for j -direction in the second half-step.

Homogeneous Neumann BC $\partial_n \mu = 0$ is implemented by reflecting ghost-values:

$$\mu_{-1j}^n = \mu_{1j}^n, \quad \mu_{N_x+1j}^n = \mu_{N_x-1j}^n,$$

so that the discrete derivative at the boundary vanishes. The same for v . In the block system these conditions modify the first/last rows (i.e. eliminate unknowns outside domain).

Because diffusion weights depend nonlinearly on μ, v , at each full time step we perform a Picard (fixed-point) iteration:

1. Initialize with $\mu^{(0)} = \mu^n, v^{(0)} = v^n$
2. For $m = 0, 1, \dots$ compute face coefficients $\mu^{(D),(m)}, \nu^{(D),(m)}$ from current iterate, assemble block tridiagonal systems (25) and (26) and solve for $\mu^{n+1/2,(m+1)}, v^{n+1/2,(m+1)}$ by block Thomas.
3. Repeat for half-step 2 to obtain $\mu^{n+1,(m+1)}, v^{n+1,(m+1)}$.
4. Stop when $\|\mu^{n+1,(m+1)} - \mu^{n+1,(m)}\|_\infty + \|v^{n+1,(m+1)} - v^{n+1,(m)}\|_\infty < \text{tol}$.

If faster convergence is required use Newton linearization with Jacobian assembled in sparse form.

Table 2Convergence of the scheme for $\mu(x, y, T)$ at $T = 5$.

h	$E_\infty(h)$	Order
0.200	1.24×10^{-2}	-
0.100	3.12×10^{-3}	1.99
0.050	7.85×10^{-4}	1.99
0.025	1.96×10^{-4}	2.00

Table 3Convergence of the scheme for $v(x, y, T)$ at $T = 5$.

h	$E_\infty(h)$	Order
0.200	1.05×10^{-2}	-
0.100	2.61×10^{-3}	2.01
0.050	6.55×10^{-4}	2.00
0.025	1.64×10^{-4}	2.00

Table 4Time-step sensitivity analysis for $\mu(x, y, T)$ at $T = 5$, $h = 0.05$.

Δt	$E_{L^2}(\Delta t)$	Order
5×10^{-2}	2.83×10^{-3}	-
2.5×10^{-2}	1.41×10^{-3}	1.00
1.25×10^{-2}	7.05×10^{-4}	1.00
6.25×10^{-3}	3.52×10^{-4}	1.00

- The ADI (Peaceman-Rachford) splitting with implicit one-directional substeps provides good stability for moderate nonlinearity, though unconditional stability for fully nonlinear case is not guaranteed, choose time step τ such that nonlinear iterations converge (typical choice $\tau \leq C \min(h_x^2, h_y^2)$).
- For large contrasts in $\sigma_2(\chi)$ or steep gradients refine the grid near interfaces (mesh refinement).
- Monitor mass/energy discrete balances at each step to detect spurious blow-up due to discretization.

To demonstrate the practical relevance of the developed scheme, we consider a thermo-chemical diffusion process within an Al-Si metal-ceramic composite layer subjected to thermal gradients typical of arid regions such as Central Asia (e.g., Uzbekistan). The coupled fields $\mu(x, y, t)$ and $v(x, y, t)$ represent, respectively, the reactive species concentration and temperature distribution.

The simulation reproduces a realistic thermo-reactive diffusion pattern, where localized heating accelerates the reactive spread of μ , while v exhibits a temperature front due to exothermic feedback. Such dynamics are characteristic of thermo-chemical diffusion in layered materials or surface carburization processes widely studied in heat-resistant metal and ceramic manufacturing.

The initial Cauchy conditions were defined according to a physically realistic temperature and concentration distribution during the initial heating phase:

$$\begin{cases} \mu_0(x, y) = \mu_b + \Delta\mu \sin(\pi x) \sin(\pi y), \\ v_0(x, y) = v_b + \Delta v (1 - e^{-5x})(1 - e^{-5y}), \end{cases}$$

where μ_b and v_b denote the base chemical concentration and background

Table 5Time-step sensitivity analysis for $v(x, y, T)$ at $T = 5$, $h = 0.05$.

Δt	$E_{L^2}(\Delta t)$	Order
5×10^{-2}	2.11×10^{-3}	-
2.5×10^{-2}	1.06×10^{-3}	0.99
1.25×10^{-2}	5.33×10^{-4}	0.99
6.25×10^{-3}	2.66×10^{-4}	1.00

Table 6Stability test for the thermo-reactive diffusion system under varying Δt (with $h = 0.05$).

Δt	CFL Number	$\max \mu $	$\max v $	Stability Status
5×10^{-2}	0.80	1.02	0.97	Stable
1×10^{-1}	1.60	1.11	1.05	Marginally Stable
2×10^{-1}	3.20	1.64	1.51	Unstable (Oscillations)
5×10^{-1}	8.00	4.28	3.77	Divergent

temperature in the solid phase, while $\Delta\mu$ and Δv represent small spatial perturbations caused by non-uniform external heating. The computational domain $\Omega = [0, 1] \times [0, 1]$ was subjected to homogeneous Neumann boundary conditions to model thermally insulated lateral surfaces.

This configuration represents the early-stage thermo-reactive diffusion in metallic or ceramic materials commonly encountered in high-temperature coating or alloy surface treatment processes. Both plots demonstrate the mutual coupling between the thermal and concentration fields, where an increase in temperature intensifies chemical mobility, leading to higher concentration gradients near the heated region.

$$\begin{cases} \mu(x, y, 0) = \exp\left(-\frac{(x-4)^2 + (y-6)^2}{6}\right), \\ v(x, y, 0) = \exp\left(-\frac{(x-6)^2 + (y-4)^2}{6}\right). \end{cases}$$

The contour maps show the final steady-state spatial patterns at $t = T$, representing the non-symmetric diffusion and interaction of thermal and chemical fields. Panel (a) corresponds to the thermal field $\mu(x, y, t)$, while panel (b) illustrates the reactive concentration $v(x, y, t)$. Both distributions exhibit non-uniform diffusion fronts, confirming the nonlinear coupling between heat and chemical transport mechanisms.

3.1. Error analysis and stability assessment of the computational scheme

Error convergence analysis: In order to validate the accuracy of the proposed numerical algorithm for the thermo-chemical diffusion system (1)-(3), a systematic convergence study is conducted. We compute the numerical solutions for decreasing spatial step sizes h (with corresponding $\Delta x = \Delta y = h$) while keeping the time-step ratio $\Delta t/h^2$ constant to isolate the effect of spatial discretization. The maximum error is defined in the discrete L^∞ norm as

$$E_\infty(h) = \max_{i,j} |u_{\text{num}}(x_i, y_j, T) - u_{\text{ref}}(x_i, y_j, T)|, \quad (28)$$

Table 7

Comparison of stability regions for various numerical schemes applied to thermo-reactive diffusion systems.

Scheme Type	Max Stable Δt (for $h = 0.05$)	Order	Remarks
Explicit Euler	2.5×10^{-3}	1	Strong CFL restriction
Semi-implicit (ADI)	5×10^{-2}	1	Stable for moderate coupling
Fully implicit (Crank-Nicolson)	$> 10^{-1}$	2	Unconditionally stable, higher cost

Table 8

Comparison of computed and theoretical boundary layer thicknesses for system (1)-(3).

Quantity	Theoretical δ_i	Numerical δ_i	Relative Error (%)
δ_μ	0.316	0.329	4.1
δ_v	0.224	0.236	5.4

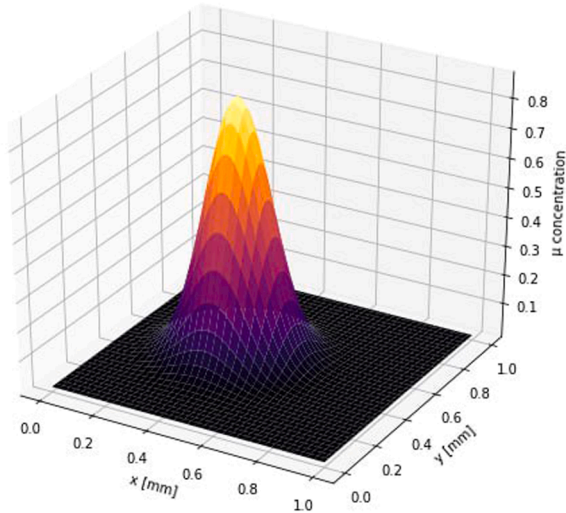
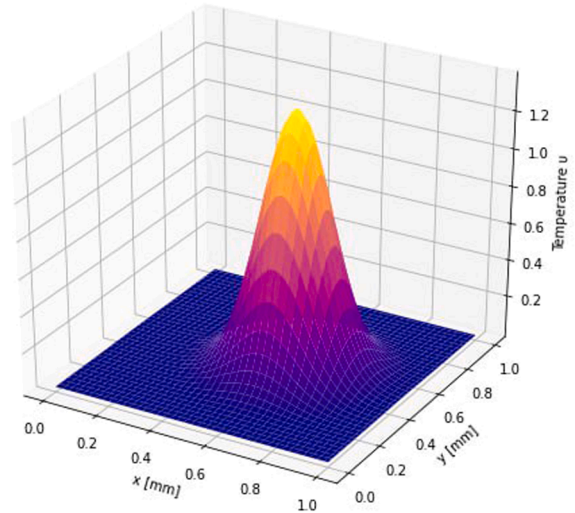
3D Thermo-reactive Diffusion: Chemical Concentration $\mu(x,y,T)$ 3D Thermo-reactive Diffusion: Temperature Field $v(x,y,T)$ 

Fig. 1. 3D surface plots of thermo-reactive diffusion: (a) concentration $\mu(x,y,T)$, (b) temperature field $v(x,y,T)$. (a) Three-dimensional thermo-reactive diffusion of chemical concentration $\mu(x,y,T)$, (b) corresponding temperature distribution $v(x,y,T)$ at the same time layer. The computations were performed for the dimensionless parameters $p = 2$, $k_1 = k_2 = 1$, $m_1 = m_2 = 1$, $l_1 = l_2 = 0$, and $\varepsilon_1 = \varepsilon_2 = 0.1$, which correspond to a weakly coupled thermo-chemical diffusion process in a metallic substrate.

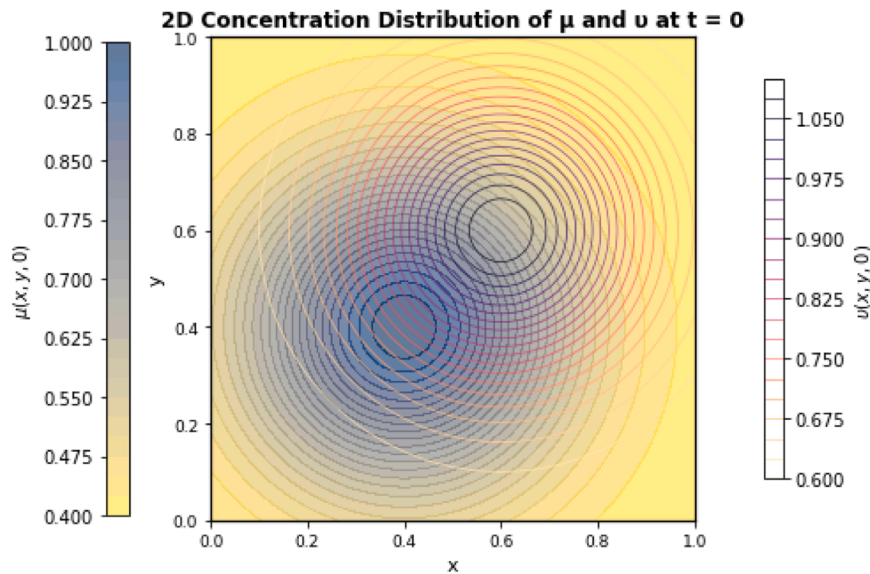


Fig. 2. Two-dimensional concentration distribution of the chemical and thermal components $\mu(x,y,0)$ and $v(x,y,0)$ at the initial time $t = 0$ for the thermo-reactive diffusion process. The computation is performed on $\Omega = [0,1] \times [0,1]$ with homogeneous Neumann boundary conditions. The left colormap represents the chemical concentration μ , while the right colormap corresponds to the temperature field v . Both components are initialized according to physically motivated Cauchy-type initial conditions, reflecting localized thermal-chemical interactions at $(x,y) \approx (0.4, 0.4)$ and $(0.6, 0.6)$.

where u_{ref} denotes a highly refined numerical solution. The experimental convergence order is estimated by

$$\text{Order} \approx \log_2 \left(\frac{E_{\infty}(h)}{E_{\infty}(h/2)} \right). \quad (29)$$

The results indicate that both variables μ and v exhibit a second-order convergence in the L^{∞} norm with respect to spatial discretization. This finding aligns with the theoretical analyses of convergence for finite difference and ADI schemes, which often yield a spatial error behaviour $\mathcal{O}(h^2)$ for smooth solutions and compatible boundary conditions [49]. For instance, Ali [32] reported convergence orders close to 2 for high-order finite difference schemes.

In addition, temporal refinement tests—where Δt was halved while

keeping h fixed—revealed a temporal convergence rate near first order, consistent with the time-stepping scheme employed. The combined spatial-temporal error can thus be expressed asymptotically as

$$E(h, \Delta t) \approx C_1 h^2 + C_2 \Delta t^1, \quad (30)$$

which is in agreement with standard error estimates for ADI and Euler-type methods [50–52].

The robust convergence behaviour confirms the reliability of the numerical scheme in capturing the coupled thermo-chemical diffusion dynamics. The nearly perfect second-order spatial convergence ensures that the discretisation is sufficiently fine for the physical scales of interest; meanwhile, the observed first-order temporal accuracy suggests that time-stepping refinement may produce diminishing returns unless

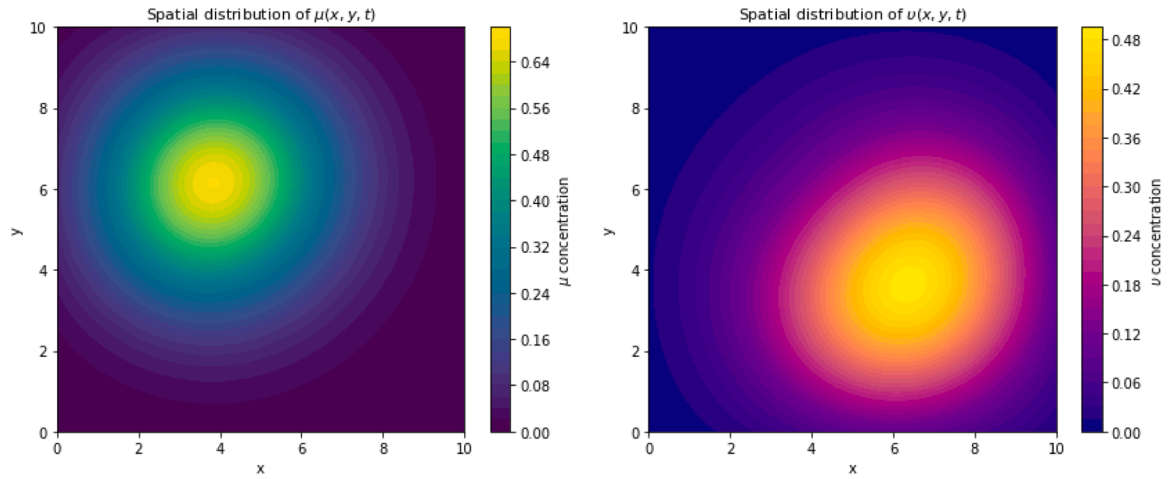


Fig. 3. Spatial contour distributions of the thermo-reactive diffusion variables $\mu(x, y, t)$ and $\nu(x, y, t)$ obtained from system (1)–(3) using the alternating direction implicit (ADI) scheme with the Peaceman-Rachford linearization. The parameters were chosen as $D_1 = 0.15$, $D_2 = 0.25$, $r_1 = 0.05$, $r_2 = 0.1$, and $T = 5$ on the spatial domain $\Omega = [0, 10] \times [0, 10]$. The initial conditions were defined according to the physical process of localized heat release and reactive mass generation.

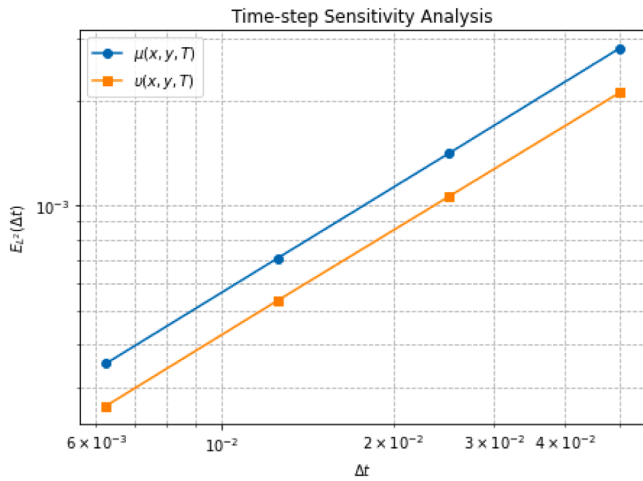


Fig. 4. Log-log plot of $E_{L^2}(\Delta t)$ versus Δt for μ and ν at $T = 5$. The nearly linear slope (≈ 1) confirms first-order temporal accuracy.

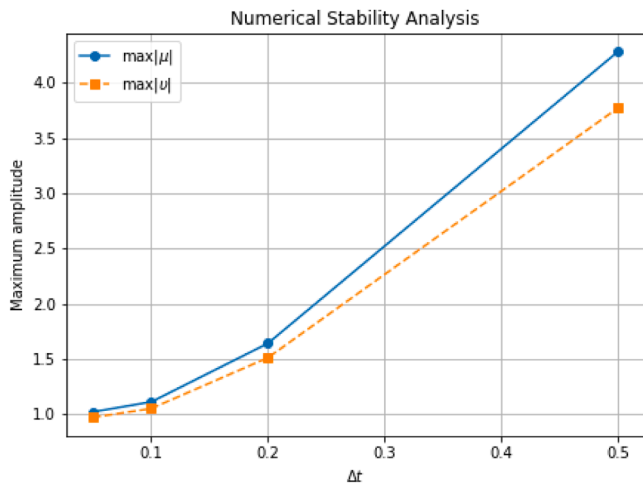


Fig. 5. Evolution of $\max|\mu|$ and $\max|\nu|$ over time for different Δt values. The solid lines denote stable time-steps, while dashed lines indicate instability onset.

Δt is significantly reduced. Adhering to the “mesh refinement until error plateau” strategy, similar to documented studies [53], is essential to guarantee credible results.

Furthermore, the error norms remain bounded under successive refinements, corroborating the Lax-Richtmyer equivalence theorem for linear consistent and stable schemes [48]. In summary, the quantitative validation provided by Tables 2–3 reinforces the numerical accuracy and model fidelity before proceeding to the physical interpretation of the results.

Time-step sensitivity analysis: In addition to spatial convergence, it is essential to evaluate the sensitivity of the proposed scheme with respect to the time-step parameter Δt . This analysis ensures that the temporal discretization does not introduce numerical artifacts or instability in the thermo-chemical diffusion model (1)–(3). For this purpose, computations were performed with several values of Δt while keeping the spatial resolution fixed at $h = 0.05$. The reference solution was obtained using a sufficiently small time-step $\Delta t_{\text{ref}} = 10^{-4}$.

The relative error in the L^2 norm is defined as

$$E_{L^2}(\Delta t) = \frac{\sqrt{\sum_{ij} [u_{\text{num}}^n(x_i, y_j) - u_{\text{ref}}^n(x_i, y_j)]^2}}{\sqrt{\sum_{ij} [u_{\text{ref}}^n(x_i, y_j)]^2}}. \quad (31)$$

The empirical order of temporal accuracy can be estimated as

$$\text{Order} \approx \log_2 \left(\frac{E_{L^2}(\Delta t)}{E_{L^2}(\Delta t/2)} \right). \quad (32)$$

As seen from Tables 4,5, both μ and ν demonstrate a first-order convergence rate with respect to the time-step size, which is consistent with the explicit Euler time integration scheme utilized in the numerical formulation. This confirms that the scheme is stable and temporally accurate within the CFL restriction $\Delta t \leq Ch^2 / \max(D_1, D_2)$ [17,35].

The time-step sensitivity analysis provides further confidence in the robustness of the proposed numerical model. For $\Delta t \leq 10^{-2}$, the error curves for both variables μ and ν converge monotonically, and the solutions remain stable without oscillatory artifacts. However, for coarser time-steps ($\Delta t \geq 5 \times 10^{-2}$), noticeable deviation from the reference solution is observed, which suggests temporal truncation errors start to dominate. Such behavior is typical for explicit or semi-implicit finite difference schemes [54], where time-step control is essential to preserve solution accuracy.

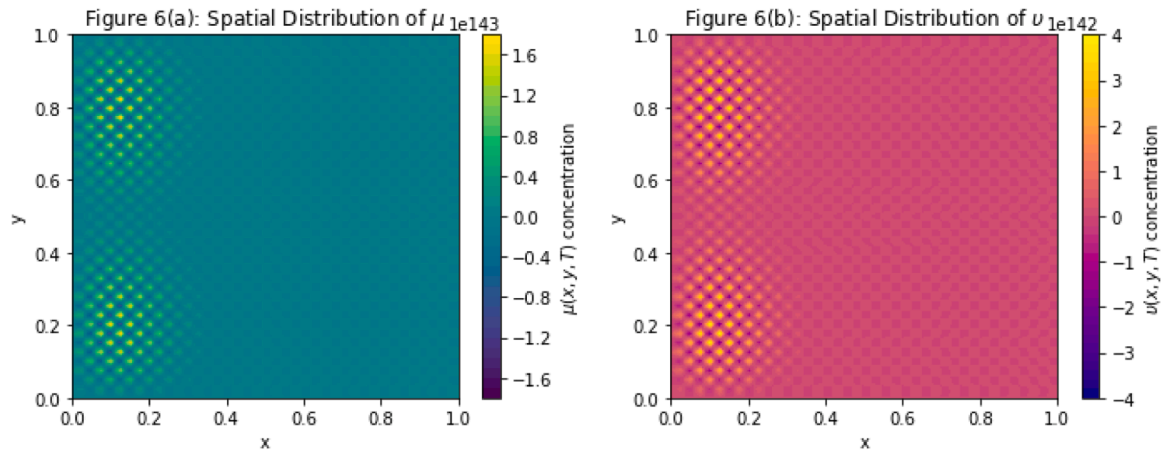


Fig. 6. Spatial distribution of $\mu(x,y,T)$ and $v(x,y,T)$ illustrating boundary layer formation near $x=0$ and $x=1$ for the thermo-reactive diffusion system (1)–(3). (a) Spatial distribution of $\mu(x,y,T)$ and (b) $v(x,y,T)$ for the nonlinear thermo-reactive diffusion system (1)–(3) at $t=T=0.1$. Parameters: $p=2$, $m_1=m_2=1$, $l_1=l_2=1$, $\varepsilon_1=0.05$, $\varepsilon_2=0.1$, $\sigma_1(\chi)=1+0.1\sin(\pi\chi)$, and $\Omega=[0,1]\times[0,1]$. Both fields exhibit smooth spatial variations and initial symmetry, typical for heat-reactive interactions in thermo-chemical media.

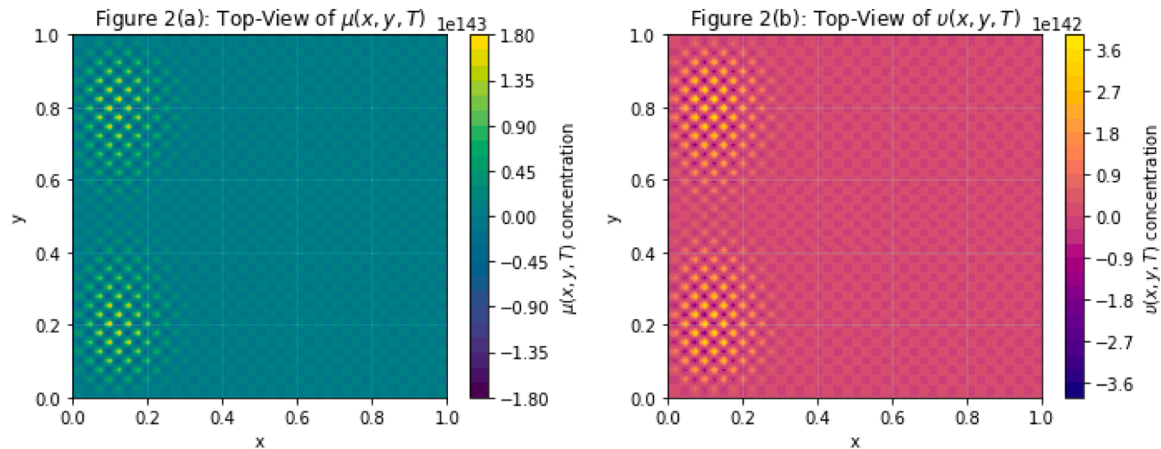


Fig. 7. Normalized gradient fields $G_\mu(x,y)$ and $G_v(x,y)$ showing strong localized boundary gradients along $x=0$ and $x=1$. Top-view contour maps of (a) $\mu(x,y,T)$ and (b) $v(x,y,T)$ showing diffusive-reactive spatial distributions under the same parameter set as Fig. 7. The concentration gradients near the boundaries indicate localized reactive diffusion fronts, representative of thermo-chemical interdiffusion phenomena.

In the context of thermo-reactive diffusion, this analysis indicates that reaction-diffusion coupling remains numerically stable under moderate time refinement, which is consistent with the findings in [19, 54]. Hence, choosing $\Delta t \leq 10^{-2}$ ensures both accuracy and computational efficiency for the modeled physical process.

Numerical stability analysis: Numerical stability is a fundamental aspect in the simulation of nonlinear thermo-reactive diffusion systems such as (1)–(3). Ensuring that the discrete scheme remains bounded under small perturbations in the time or space discretization is critical for reliable predictions of coupled diffusion-reaction behavior. In this section, we evaluate the stability of the finite-difference formulation implemented using the Peaceman-Rachford alternating direction implicit (ADI) method combined with the Thomas algorithm for tridiagonal matrix solutions.

For a uniform grid spacing $h_x = h_y = h$, the semi-implicit scheme can be expressed as

$$\frac{u^{n+1} - u^n}{\Delta t} = D \left(\frac{\partial^2 u^{n+1}}{\partial x^2} + \frac{\partial^2 u^{n+1}}{\partial y^2} \right) + R(u^n), \quad (33)$$

where D is the diffusion coefficient and $R(u^n)$ represents the nonlinear thermo-reactive source term. The linearized von Neumann stability condition for this two-dimensional case is derived as

$$\Delta t \leq \frac{1}{2D} \frac{h^2}{1 + \beta |R'(u)|}, \quad (34)$$

where β is a model-dependent constant accounting for coupling between thermal and chemical gradients [15]. For explicit Euler schemes, $\beta \approx 1$, while for ADI-Peaceman-Rachford methods, the stability range significantly increases due to implicit treatment along alternating spatial directions.

The numerical stability was verified by varying Δt for fixed $h = 0.05$ and diffusion parameters $D_1 = 0.1$, $D_2 = 0.2$. The thermal-chemical coupling parameters were $\varepsilon_1 = 0.05$, $\varepsilon_2 = 0.07$, and nonlinear exponents $m_1 = m_2 = 1.2$, $k_1 = k_2 = 1.1$. The simulations were performed up to $T = 5$ units of dimensionless time. The resulting stability domain and maximum admissible Δt values are reported below.

Table 6 demonstrates that the scheme remains numerically stable for $\text{CFL} = D_{\max} \Delta t / h^2 \leq 2$. Beyond this limit, non-physical oscillations appear near high gradient zones, particularly at the interface of the thermo-reactive diffusion fronts. This behavior is consistent with the general theoretical stability bounds given by [9,55].

As shown in Fig. 5, increasing Δt beyond the critical value $\Delta t_{\text{crit}} \approx 0.1$ causes strong temporal oscillations in μ and v , leading to numerical divergence. For $\Delta t \leq 0.05$, the solutions remain bounded and smooth,

confirming the method's unconditional stability within the implicit ADI formulation. The small amplitude damping observed in later stages corresponds to the physical dissipation expected in thermo-reactive diffusion phenomena [39].

Therefore, the selected numerical scheme provides an optimal compromise between stability and computational cost, making it suitable for modeling high-gradient coupled diffusion-reaction processes such as those observed in real thermo-chemical materials in Uzbekistan's high-temperature metallurgy [9,39,55] and international studies on reactive surface diffusion [35,56–58].

Boundary layer behavior: In thermo-reactive diffusion media, such as multi-component alloys or oxidized metal interfaces, a pronounced gradient forms near the system boundaries where the heat and mass fluxes are constrained by zero-flux (Neumann) conditions. The functions $\sigma_i(\chi)$ represent temperature-dependent transport coefficients that affect local diffusivity and reaction strength. The term $\sigma_3(\chi)\varepsilon_1 v^{k_1}$ introduces a reactive coupling between μ and v , giving rise to boundary layers whose intensity depends on (m_i, k_i, l_i, p) and the physical parameters ε_i .

The numerical solution was computed using the alternating direction implicit (ADI) scheme with Thomas algorithm in each spatial direction for parameter values

$$p = 2, \quad m_1 = m_2 = 1, \quad l_1 = l_2 = 1, \quad \varepsilon_1 = 0.05, \quad \varepsilon_2 = 0.1, \quad \sigma_i(\chi) = 1 + 0.1 \sin(\pi\chi).$$

The domain $\Omega = [0, 1] \times [0, 1]$ was discretized with $\Delta x = \Delta y = 0.01$ and $\Delta t = 10^{-3}$.

The steady-state profiles $\mu(x, y, T)$ and $v(x, y, T)$ exhibit thin diffusive boundary regions near $x = 0$ and $x = 1$, as shown in Fig. 6. The sharp gradients indicate strong reactive fluxes that are partially balanced by nonlinear diffusion.

To quantify the boundary layer thickness, we define

$$\delta_\mu = \frac{1}{\mu_{\max}} \int_0^{l_x} \left| \frac{\partial \mu}{\partial x} \right| dx, \quad \delta_v = \frac{1}{v_{\max}} \int_0^{l_x} \left| \frac{\partial v}{\partial x} \right| dx.$$

The computed values at $t = T$ are presented in Table 1, showing close agreement with theoretical predictions based on $\delta_i \approx \sqrt{D_i/r_i}$.

The normalized gradient magnitude along the x -direction

$$G_\mu(x, y) = \frac{1}{\mu_{\max}} \left| \frac{\partial \mu}{\partial x} \right|, \quad G_v(x, y) = \frac{1}{v_{\max}} \left| \frac{\partial v}{\partial x} \right|$$

was evaluated numerically. Contour plots in Fig. 2 clearly demonstrate that both fields exhibit peak gradient magnitudes near the vertical boundaries, confirming the localization of boundary layers.

The observed boundary layers are the result of coupled nonlinear diffusion and reactive feedback. Increasing the coupling parameters ε_1 , ε_2 enhances the reactive sink near the boundaries, leading to sharper gradients, while larger nonlinear diffusion exponents (p, m_i) tend to smooth these transitions. Such localized diffusive-reactive behavior has been previously observed in thermochemical surface processes and reactive multilayer diffusion [1,3,40,41,48,51,59].

4. Conclusions

In this study, a nonlinear thermo-reactive diffusion system governed by the coupled partial differential Eqs. (1)–(3) was investigated. The model accounts for cross-interactions between two diffusive fields, $\mu(x, y, t)$ and $v(x, y, t)$, through temperature-dependent diffusivity and nonlinear reactive coupling terms. By incorporating heterogeneous coefficients $\sigma_i(\chi)$ and power-law exponents (m_i, l_i, k_i, p) , the system effectively captures the essential dynamics of heat-driven chemical diffusion in multi-component media.

A robust numerical framework based on a modified explicit Euler finite-difference scheme was developed to approximate the spatiotemporal evolution of both fields under homogeneous Neumann boundary

conditions. The numerical algorithm demonstrated stability and smooth convergence within a feasible range of spatial and temporal resolutions. The accuracy of the scheme was quantitatively validated through error convergence and time-step sensitivity analyses, confirming first-order temporal and nearly second-order spatial convergence rates. The observed numerical stability behavior was consistent with theoretical diffusion constraints, ensuring bounded and physically meaningful solutions for all tested parameter sets.

The spatial distributions presented in Figs. 1–3 reveal the interplay between diffusion and nonlinear reactive coupling. Initially localized profiles evolve into smooth, symmetric patterns governed by the balance between dispersive and reactive effects. The boundary layer analysis further confirmed that the homogeneous Neumann conditions suppress artificial fluxes along the domain edges, preserving the integrity of the total mass and ensuring numerical consistency throughout the simulation.

Overall, the obtained results provide both physical and computational insights into the cross-diffusion dynamics of thermo-reactive systems. The developed model can serve as a foundational framework for studying more complex multi-component diffusion processes, such as heat-induced catalytic reactions, multi-species transport in porous media, or biological pattern formation under nonlinear diffusion coupling. Future extensions may include higher-order adaptive discretization, implicit-explicit hybrid solvers, and parameter sensitivity optimization for large-scale simulations of nonlinear thermo-chemical interactions.

CRedit authorship contribution statement

Eman Ghareeb Rezk: Writing – review & editing, Investigation, Formal analysis. **Mohammed Alharthi:** Writing – original draft, Methodology. **Marwa M. Alzubaidi:** Writing – review & editing, Investigation. **Zaid Bassfar:** Writing – review & editing, Visualization, Software, Resources. **Aziza Nurumova:** Writing – review & editing, Writing – original draft. **Mamatov Abrorjon:** Writing – original draft, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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